Group-1

Bangladesh University of Engineering and Technology

PROJECT PROPOSAL ON
A COMBINED HALL CANTEEN AT BUET



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CONSULTANCY

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EXECUTIVE SUMMARY

Canteen facility is one the most important components regarding the welfare of the general students of BUET. Recently the issue regarding degrading quality of foods and service of canteen staffs has been the highlighted issue. Therefore, the urge to establish a hall canteen which will also ensure the safety and security for female students. A convenient location is selected through performing feasibility study and Ahsan Ullah Hall Canteen delivers the best convenience considering suitability of location and surroundings. Past canteen structure will be replaced by a more sustainable two storied reinforced concrete structure. These 5000 square feet of total canteen space will provide the best modern restaurant facilities, but capacity, cleanliness and quality of food will be emphasized. A dedicated female section will provide the upmost comfort and privacy. The timeline of the entire project will be divided into several sequence of activities. Expected schedule of the project is from 09/01/2023 to 09/09/2023. For the completion of the project, funding of around 40 million taka is been estimated.

INTRODUCTION

The purpose of the project is to develop a combined hall canteen that would serve as an eating place with much more comfort and convenience than the existing hall canteens. Current hall canteens lack in providing good quality food at a student greater comfort, especially for girls. The project would provide a convenient and accessible location for the massive number of students offering a variety of food, and recreational features, and would replace the many food locations on the campus which will improve resources and efficiency.

After demolishing the existing Ahsan Ullah and Nazrul hall canteen, the project would build a two-story modern canteen complex with the motive of student convenience by adding special features targeted for the best use of space and funding. A fantastic project team of consultants and engineers would guide the project throughout the entire process.

PROJECT DESCRIPTION

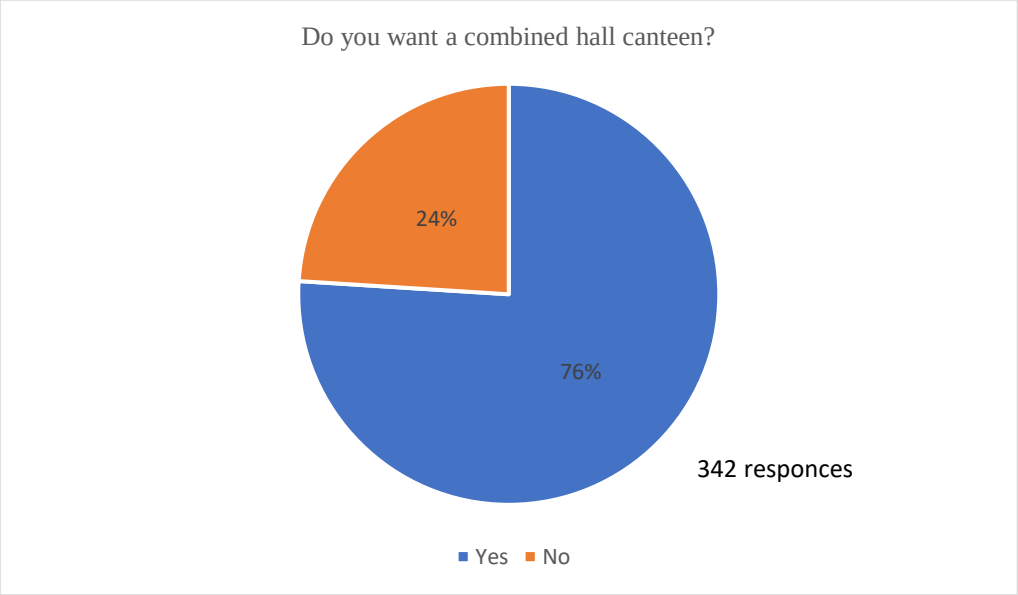
OBJECTIVE

Our objective is to

- Design a combined hall canteen within one year.
- Reducing or eliminating the need for multiple food service locations on campus, can save resources and improve efficiency.
- Providing a convenient and accessible location for students or employees to purchase food and drinks.
- Offering a variety of meal options to meet different dietary needs and preferences.
- Promoting healthy eating habits through the provision of nutritious food options.
- Generating revenue for the school or organization through food sales.
- Creating a welcoming and inclusive atmosphere for all members of the community.
- Improving the overall student or employee experience by providing a central gathering place.
- Enhancing the overall aesthetic and functionality of the building or facility.

METHODOLOGY

To assess the demand for a combined hall canteen, we conducted an online survey. Almost 76% current students of BUET agreed that we do need one in BUET.



FEASIBILITY STUDY

Ahsan Ullah Hall Canteen is the suggested location for the combined hall canteen. It is on Ahsan Ullah Hall's western side. There is not enough room within BUET. A combined hall canteen must be situated such that it is both conveniently accessible and not too far from the campus and residential halls. We have chosen to raze Ahsan Ullah Hall Canteen and Nazrul Hall Canteen for this project in favor of constructing a new, two-story structure for a combined hall canteen. The residential halls and main campus are not too distant from this spot, which is widely known among the students. A five thousand square foot structure might be constructed on the available land.

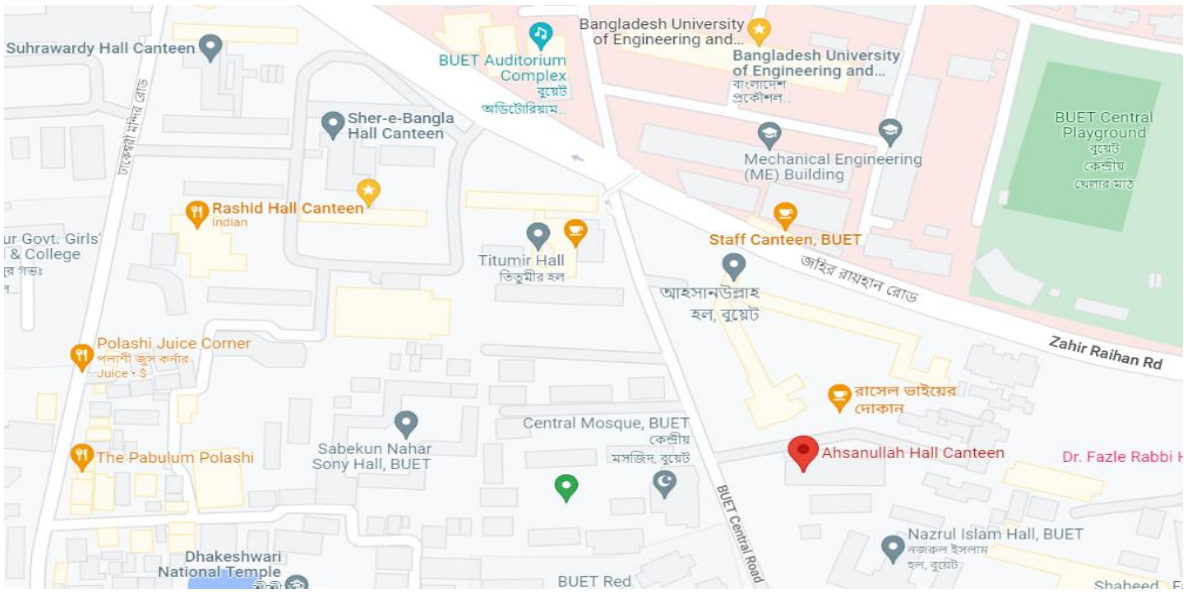


Figure 1: Location of Ahsan Ullah Hall Canteen

FACILITIES

Structural:

Glass rooftop allows –

- Energy efficiency: drastically reducing electricity bills.
- Better illumination: allows ample natural light into the house.
- Reduced noise Increases the resale value of the house.

Capacity:

This will be a two storied building where 25 tables will be set for the ground floor, 45 tables for the first floor and 15 tables for the rooftop. In each table 6 people can easily sit. On the ground floor there will be a partition beside the food counter where the kitchen is located. Stairs will be used to transfer food from kitchen to counter.

Service:

From the capacity section we can already determine the number of waiters and manager at the counter we need for a two storied cafeteria. For 5 tables there will be one waiter allotted. So, for ground floor 5, for first floor 9 and for rooftop 5 waiters are allotted. As the kitchen is on the ground floor, foods will be served in two ways:

1. Waitering
2. Self-servicing

Both ways are carried out for the ground floor. Floor first floor and rooftop, additional beeping machine service is done to ensure timely and satisfying waitering. Four managers will be allotted for the food counter to take control over financial and overall servicing works who will work alternately in pair.

Food:

Food quality will be ensured by regular buying of fresh raw materials of cooking from market. Variety of foods will be ensured according to students preferences. To ensure this, two students' representative will be selected who will make a bridge among the students and the cafeteria authority. There will 2 vending machines in each floor consisting of soft drinks and packaged food items. In addition to all these, delivery system to all halls and digital payment with apps will be also available for flexibility.

Cleanliness:

This is one of most sensitive issue that needs to be handled carefully. So to ensure overall cleanliness four people will be given the authority. Sanitization, use of masks of waiters and chefs, properly cleaning cooking materials and serving utensils, washroom cleaning need to be ensured on a regular basis.

Recreation:

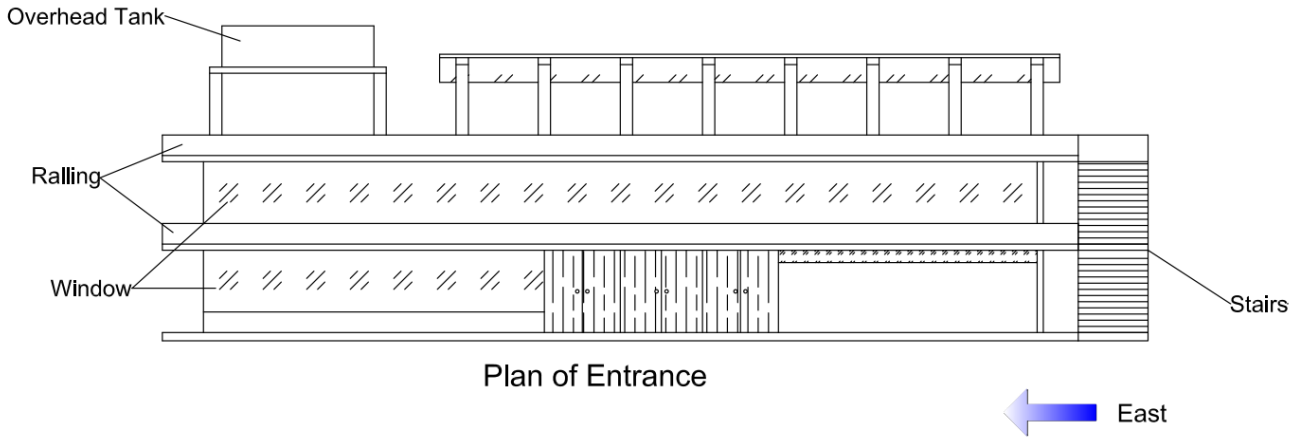
Ground floor will have an arrangement of television display so that people can enjoy events like football, cricket matches while eating all together. There will be a corner with a bookshelf consisting of comics, novels etc. too if anyone wants to enjoy reading while eating. On rooftop part there will be two video games settings and a stage arrangement where many events like jamming session, cultural nights can be held.

Miscellaneous:

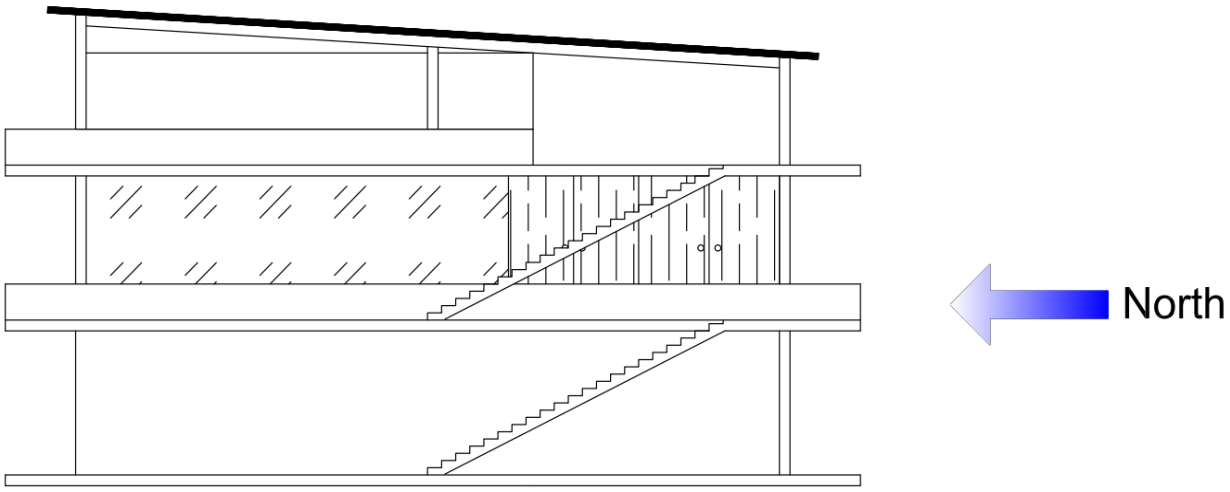
In the first floor there will a separate section for the female students to make a comfortable space for them. In the rooftop there will be a smoking zone too.

DESIGN & STRATEGY

Ahsan Ullah and Nazrul Hall's cafeteria is dated. Another storey would have increased the space, but the canteen's construction cannot sustain it. As a result, both canteens will be destroyed, and a new, two-story structure will be built to serve as the combined hall canteen. For each level, there are 5,000 square feet of floor area. Two floors and a rooftop area will be present.

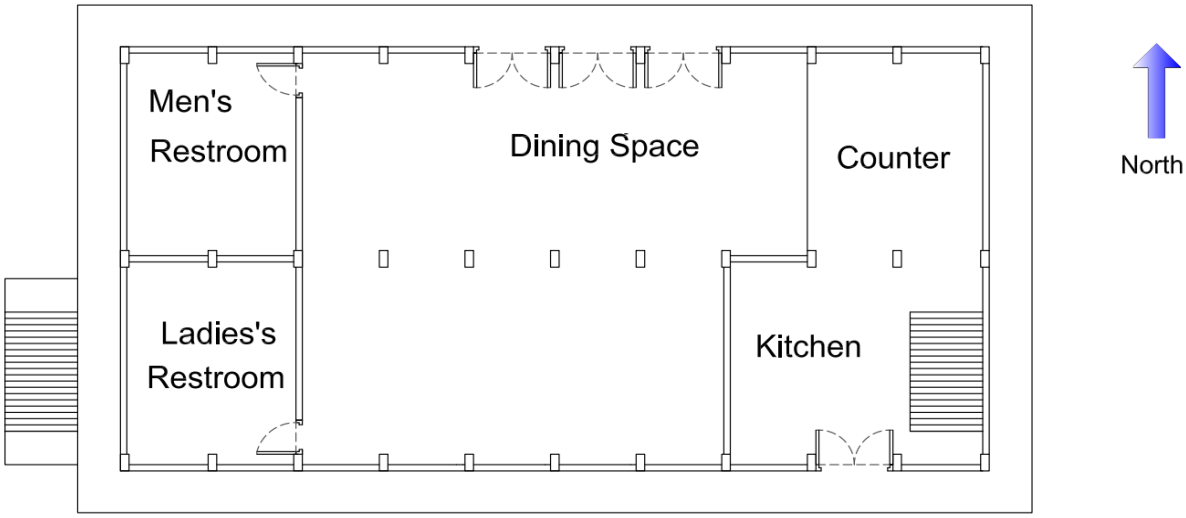


Plan of Entrance



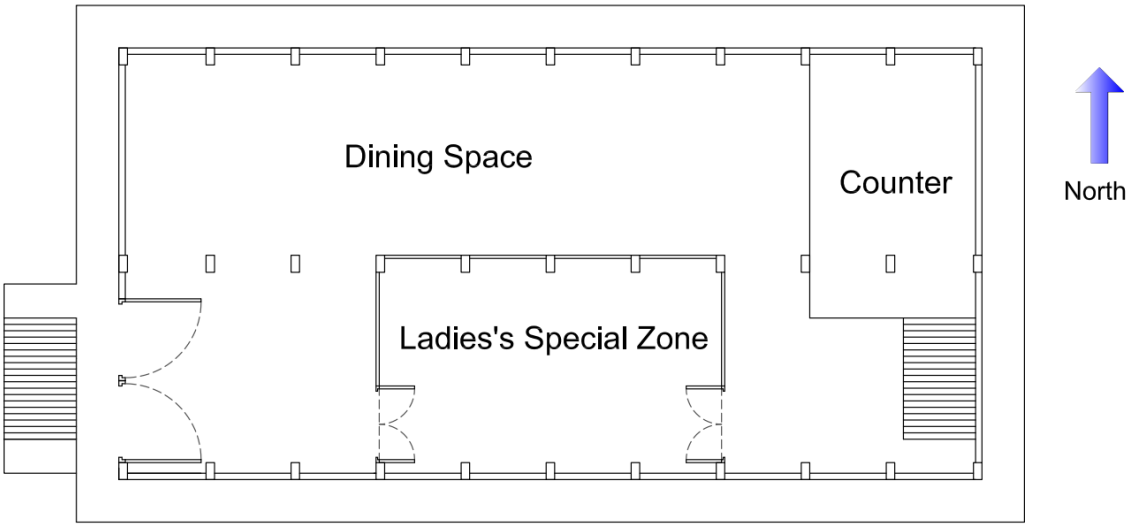
Side View

On the ground level, there will be a counter, a kitchen, a dining area, and restrooms for the students. The workers will have access to a separate staircase to bring meals to the first level. There will be a 980-square-foot kitchen. There will be 2600 square feet of eating space. The restroom will have 960 square feet of area. The remainder is for the counter. 150 persons will be seated according to plans.



Plan of Ground Floor

For the first-floor concept, we have set aside 4250 square feet for eating space, including a designated area just for women. Arrangements for seating 270 persons will be made. The remaining space will be used as a storage and food counter.



Plan of 1st Floor

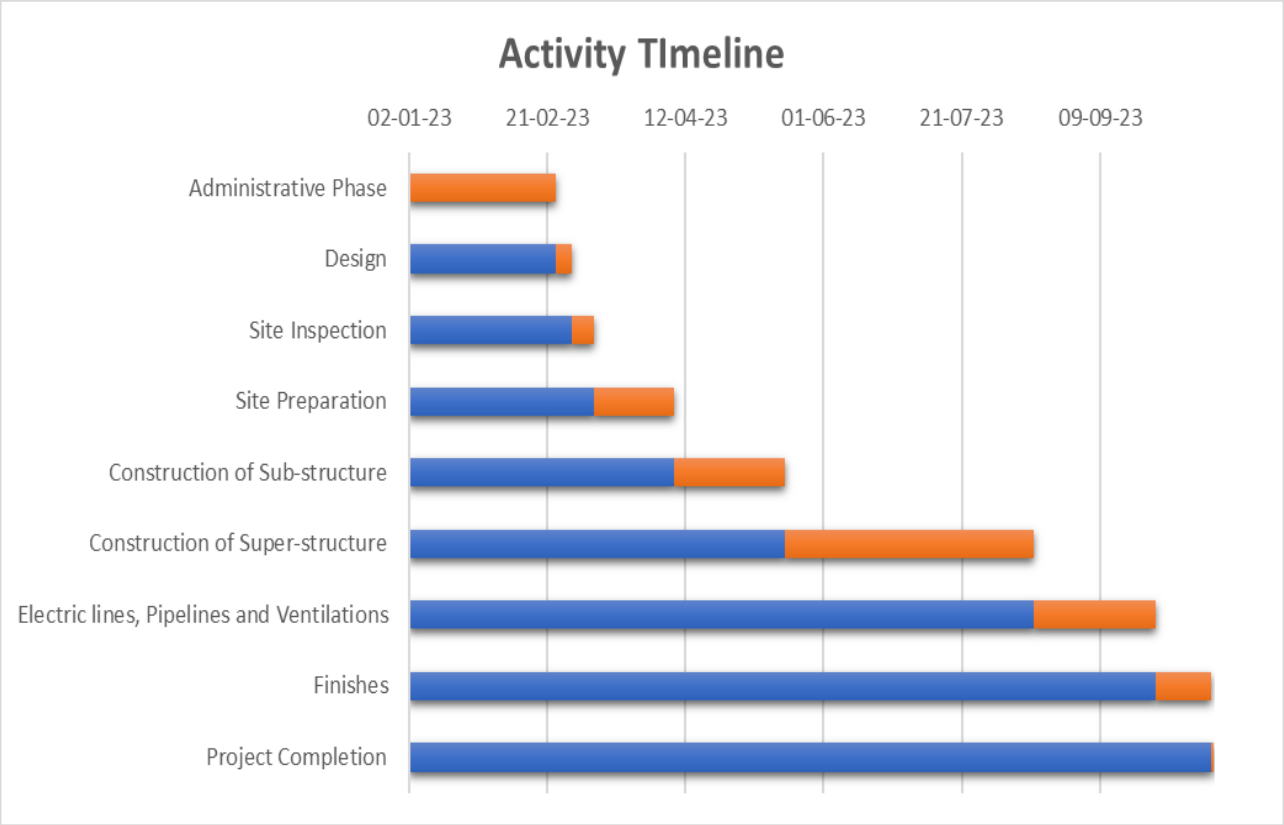
We have created a recreation facility on the roof. The pupils' performances will take place on a stage. The students are also welcome to eat here. In all save the roof, it will be open. To ensure sunshine and provide rain protection, we will utilize tinted tempered glass as the roof. For safety, there will be a railing. There will be a designated area for smokers.

ACTIVITY PLAN

Work Phase	Details		Starting Date	Completion Date	Duration Period	Phase Duration
Administrative Phase	Project Proposal & Approval		02-01-2023	09-01-2023	7days	53days
	Formation of Administrative Panels		09-01-2023	12-01-2023	3days	
	Distribution of Work		12-01-2023	15-01-2023	3day	
	Tendering		15-01-2023	24-02-2023	40days	
Design	Formation of Design Committee with architectural Department		24-02-2023	28-03-2023	4days	6days
	Design selection		28-03-2023	02-03-2023	2days	
Site Inspection	Survey the project location.		02-3-2023	05-03-2023	3days	8days
	Soil test on site		05-03-2023	10-03-2023	5days	
Site Preparation	Demolishing the existing structure		10-03-2023	17-03-2023	7days	29days
	Compaction of ground		17-03-2023	24-03-2023	7days	
	Excavation		24-03-2023	08-04-2023	15days	
Construction of Sub-structure	Foundation		08-04-2023	18-05-2023	40days	40days
	Footings					
	Grade Beam					
	BFS					
Construction of Super-structure	1st Floor	▪ Columns ▪ Beams ▪ Slabs ▪ Walls ▪ Lintels ▪ Stairs	18-05-2023	17-06-2023	30days	90days
	2nd Floor		17-06-2023	17-07-2023	30days	
	3rd Floor		17-07-2023	16-08-2023	30days	
Electric lines, Pipelines and Ventilations	Electricity	Designing Conduit Layout	16-08-2023	26-08-2023	10days	44days
		Installation of conduit, fixtures, and fitting				
		Installation of wiring				
	Plumbing		26-08-2023	02-09-2023	7days	
	Doors, Windows		02-09-2023	22-09-2023	20days	
	Ventilation	Installation of HVAC system	22-09-2023	29-09-2023	7days	

Finishes	Floor Finish (Tiles) Plastering Painting	29-09-2023	19-10-2023	20days	20days
Project Completion		19-10-2023	20-10-2023	1day	1days
Total days				291days	291days

Grant Chart of Activity Timeline



MATERIALS DESCRIPTIONS

- Concrete Block
- Reinforcements
- Stone Chips
- Sand
- 10 mm Tempered Glass
- Tiles

Working Method of the Materials

- **Concrete Block:** A concrete block is a primary construction material, providing the foundation for the development of a building. It is one of the several precast concrete products that ensure the longevity and durability of a development. Concrete masonry is truly an economical building solution. In addition to being competitively priced to other building materials, concrete masonry sets itself further apart from the competition when you factor in benefits such as energy efficiency, low maintenance, sustainability, and green building properties.

- **Reinforcement**: The strength that reinforced steel provides in support of the concrete is a force multiplier in terms of durability and protection for the interior of the pool. During the time of construction, the required reinforcements are used according to the design of the pool.

- **Stone chips:** Stone chips are small pieces of stone that are used, for example- in concreting, along with cement and sand.

- **Sand:** Sand is a granular material composed of finely divided mineral particles. Sand has various compositions but is defined by its grain size. Here we will use sand for normal constitutional purposes.

- **10 mm tempered Glass:** Tempered glass can provide durability, as its name implies, it is very resistant to changes in air temperature, and for a long time. By using this tempered glass, we can secure a more comfortable building.

- **Tiles:** There are many types of tiles that can be used to cover the walls of a canteen building. Some examples of tiles are vinyl, ceramic, and glass panes. Each type of tile has different characteristics that can make it more suitable for buildings' design and maintenance needs.

COST ESTIMATION:

Cost Estimation of Dismantling and Demolishing:

Item no	Description	Unit	Unit Rate	Total Quantity	Total Amount
1	Dismantling of brick works of thickness 250 mm or more in foundation and in plinth including any type of plaster, skirting, dado etc. by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety, maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project site area as per directions and acceptance by the Engineer-in-charge	cum	Tk. 1,127	20	Tk. 22540
2	Dismantling of brick works of thickness 75/125mm in foundation and plinth including any type of plaster, skirting, dado etc. by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety, maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge	sqm	Tk. 169	300	Tk. 50700
3	Dismantling of brick works of thickness 75/125 mm in Superstructure including any type of plaster, skirting, dado etc. by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety; maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge(Up to Ground Floor Roof)	sqm	Tk. 177	240	Tk. 42480
4	Dismantling of plain concrete made of lime or cement including any thin finishing on it by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety; maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris,	cum	Tk. 2552	12	Tk. 30624

	salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge				
5	Dismantling of brick flat soiling by manual means; cleaning the surface by soft brooming, providing provision for safety; maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge	sq m	Tk. 38	300	Tk. 11400
6	Dismantling of R.C foundation, foundation beam, grade beam including any plaster and thin finishing on it by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety; maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge	cu m	Tk. 5027	10	Tk. 50270
7	Dismantling of R.C beam, Lintel, Tie Beam, Column, Wall, Drop wall, Railing including any plaster, skirting/ dado and finishing on it by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety; maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge(up to Ground floor)	cu m	Tk. 5461	12	Tk. 65532
8	Dismantling of R.C in roof slab, sun shed, cornice including any plaster and thin finishing on it by using electro-mechanical equipment; cleaning the surface by soft brooming, providing provision for safety; maintaining tolerable sound level and minimizing dust hazard, in constrained conditions/ limited scope of work to utilize full capacity of manpower, equipment and removal of debris, salvaged materials and stacked in a safe place within the campus/ project sire area as per directions and acceptance by the Engineer-in-charge(Up to Ground Floor)	cu m	Tk. 4442	46	Tk. 204332

9	Removal of all types of debris/ extra unstable earth/ unusable sand/ mud/ sludges etc. from the site/ campus by truck/ cart in such a way that no carrying materials fall outside during carrying etc. in connection with retrofitting works only, in constrained conditions as per direction and approval of the engineer-in-charge	cu m	Tk. 435	150	Tk. 65250
	Total				Tk. 543128

Cost Estimation of the Combined Hall Canteen

Building Type : Non-residential
Type of Structure : RCC frame structure with 1:1.5:3 concrete (brick chips)
Soil Type : Bearing Capacity of Soil = 3 ksf
Foundation : Shallow foundation
Plinth Area : 465 sqm
Site : Dhaka

1. Soil Type:

	Description	Quantity/Amount		Unit Rate (Tk.)	Total Amount (Tk.)
	Soil Investigation	6	Nos.	25000	150000

2. Construction of Building

A. Foundation cost

Item no	Description	Quantity/Amount		Unit Rate (Tk.)		Total Amount (Tk.)
1.	Foundation cost (Allowable Bearing Capacity=3.0 ksf, Storey =2)	465	sqm	12724	Per sqm	5916660

B. Super Structure cost:

Item no	Description	Quantity/Amount		Unit Rate (Tk.)		Total Amount (Tk.)
1.	Ground Floor	465	sqm	26605 +821	Per sqm	12753090
2.	1st Floor	465	sqm	25246 +548	Per sqm	11994210

C. Roof Top Cost:

Item no	Description	Quantity/Amount		Unit Rate (Tk.)		Total Amount (Tk.)
1.	Rooftop RCC parapet	97	sqm	4642	Per sqm	450274
2.	Extra cost for Rooftop & LC	465	sqm	3360	Per sqm	1562400
3.	Extra cost for Roof top R.C.C water tank	2000	gal	178	Per gal	356000
4.	Glass Roof (10mm glass) + Tempering	480	sqm	300 + 250	Per sqm	264000

3. Internal Water Supply and Sanitation:

Item no	Description	Quantity/Amount		Unit Rate (Tk.)		Total Amount (Tk.)
1.	Non-residential super	400	sqm	1563	Per sqm	625200

4. Internal Electrification:

Item no	Description	Quantity/Amount		Unit Rate (Tk.)		Total Amount (Tk.)
1.	Non-residential super	400	sqm	1930	Per sqm	772000

5. Gas Connection:

Item no	Description	Quantity/Amount		Unit Rate (Tk.)		Total Amount (Tk.)
1.	Ground Floor	30	sqm	455	Per sqm	13650

Sub total		34557484
Quality Assurance, material, sample collection & test (1% of subtotal)		345574.84
Contingency (10% of Subtotal)		3455748.4
Price contingency (8%)	Physical contingency (2%)	
Grand Total		38358807.24

HUMAN RESOURCES:

Construction	
Human Resource	Number
Project Manager	1
Construction Manager	1
Architect	1
Structural Engineer	1
Mechanical Engineer	1
Electrical Engineer	1
Quantity Surveyor	3
Site Manager	1
Laborer	15
Plumber	5
Painter	5

Canteen Management	
Work Force	Number
Chef & Kitchen staff	5
Manager	4
Waiter	15
Cleaner	3

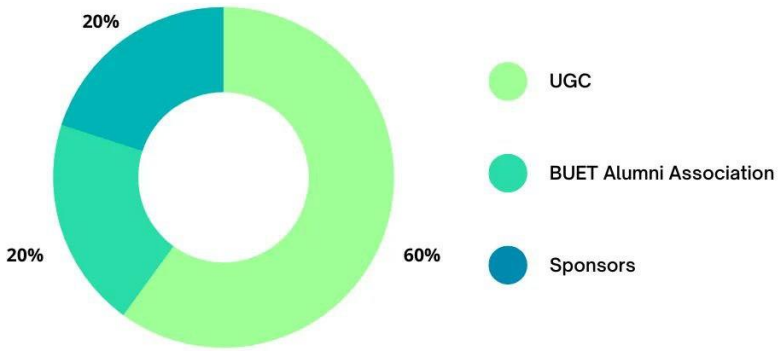
DIVISION OF FINANCIAL SOURCES:

UGC: All of Bangladesh's affiliated public, private, and national universities are regulated by the UGC. For Bangladesh's "Government-Funded Universities," it provides financial assistance. BUET is one among them. Its objective is to ensure the quality of higher education across the country. Therefore, we anticipate that this institution will manage 60% of the project's costs.

BUET Alumni Association: BUET alumni has always been enthusiastic, helpful and responsive regarding any development work, festival or donation in BUET.

Acknowledging their positive and helpful gesture, 20% of the fund is expected to be accrued from them.

Sponsorship: Rest 20% of the fund is estimated to be obtained from the sponsors like Coca-Cola, Pepsico, Igloo, Akij Foods and so on.



EVALUATION INDICATOR

The target of this project is to make a quality-enriched environment for the students in a feasible way. We are expecting improved facilities regarding food, services, and hygiene than that of the old one. We are also expecting increased engagement of students as we have focused on the recreation section too along with food.

Our target audience who will be benefitted mostly from this project is the student community. As per our target and expectations, employees and the authority working on this project will make profits too if students are benefitted accordingly.

As we are saying through this we can serve delicious and healthy food at the right price, so the students of this institution do not have to spend more money in the costly restaurants for good food. At the same time, students do not have to go out for recreation and even students from other institutions will be interested to come here. This will also save our students' time. Additionally, we can be safe by staying on our own campus.

This project will be able to implement the concept of an ideal and modern canteen in front of everyone. It will create curiosity among other universities through which the reputation of our university will be enhanced.

SWOT ANALYSIS:

SWOT analysis of Union Allied Consultancy for BUET Combined Hall Canteen Project:

Strengths:

- Strong experience in canteen design and implementation
- Knowledge of food service operations and equipment
- Understanding of local health and safety regulations
- Ability to work within budget and time constraints

Weaknesses:

- Limited experience in large-scale hall canteen projects
- Lack of expertise in incorporating sustainability features
- Dependence on external suppliers for equipment and supplies
- Potential resistance from stakeholders to change

Opportunities:

- Growing demand for cost-effective and efficient canteen solutions
- Increased focus on sustainability and eco-friendly practices
- The possibility of creating a unique and innovative design to stand out in the market
- Potential to expand services into other hall canteen projects

Threats:

- Competition from other consultancy firms with more experience in hall canteen projects
- Economic uncertainties and budget constraints
- Changes in regulations and standards affecting food service operations
- Unforeseen challenges during project implementation, such as logistical and technical difficulties.

CONCLUSION

The prime objective of this project is to establish a combined hall canteen for the students of BUET. As one of the leading educational institutions of Bangladesh, it's really a requirement to provide better quality of food. Because human health is directly influenced by the food consumption. As a result, to ensure psychological welfare which is directly related with how a student performs in his or her study it is of utter importance to provide better food. BUET Combined Hall Canteen facility is going to be a significant factor considering the best possible solution to all the existing factors. Different project activities will facilitate the students along with enhance the academic, recreational, economic perspective of BUET. Therefore, addressing to all the above factors, we anticipate the concerning authority will find this project proposals suitable and implement these ideas as soon as possible to improve the overall quality of the cafeteria.

ACKNOWLEDGEMENTS:

- Sumaiya Afrose Suma, Assistant Professor, Department of Civil Engineering, BUET
- Saidis Salekin Aninda, Lecturer, Department of Civil Engineering, BUET
- Madeha Sattar Khan, Lecturer, BUET-JIDPUS, BUET
- ENGR. Mohammad Abu Kauser, Executive Engineer (Civil), Engineering Office, BUET

REFERENCE:

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